

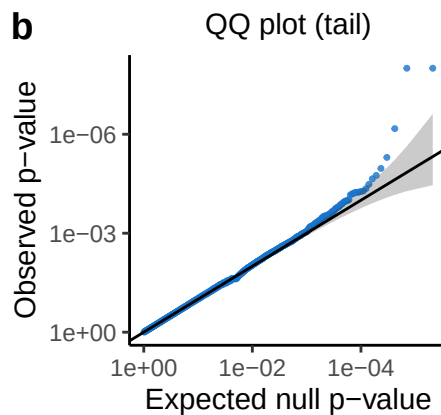
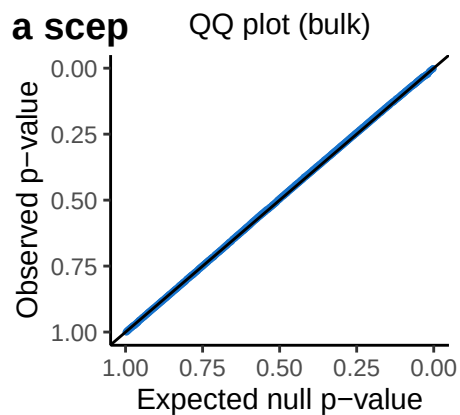
gasperini-analysis-highly-expressed

2026-03-19

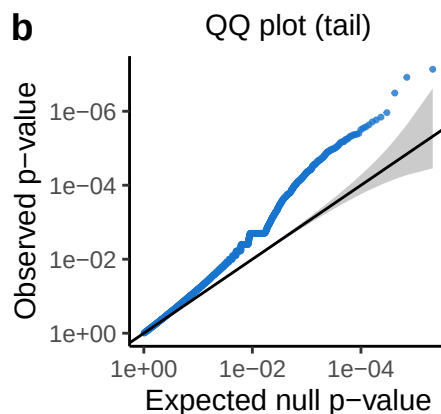
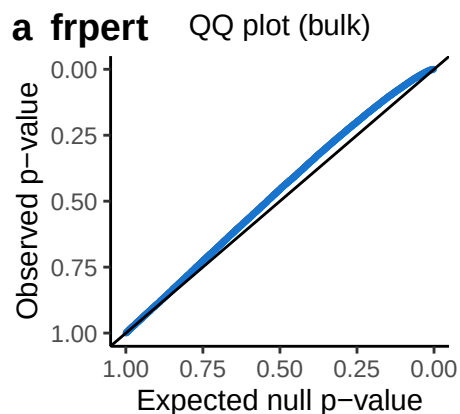
This notebook makes figures to explore the following:

analysis of positive and negative control on gasperini, only using highly expressed genes.

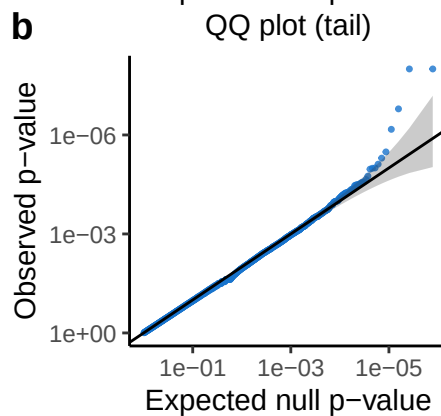
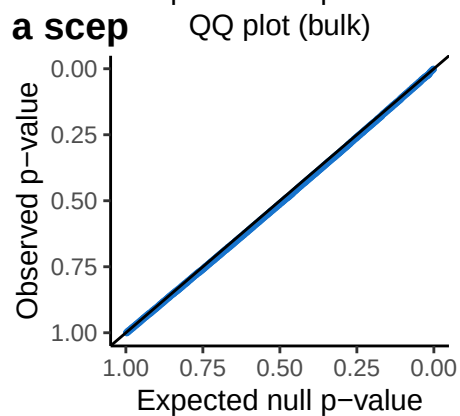
```
## Dataset: gasperini_neg_control_4000genes : #genes = ; #cells = 18011
## #guides = 101 ; #targets = 101 ; avg # guides / target: 1
## # cells per guide: median = 206 ; min = 66 ; max = 416
## # cells per target: avg = 206 ; min = 66 ; max = 416
## # guides per cell: moi (avg) = 1.145467 ; min = 1 ; max = 5
## # discovery pairs: 404000
##
## *****
## Dataset: gasperini_pos_control : #genes = 377 ; #cells = 185171
## #guides = 855 ; #targets = 378 ; avg # guides / target: 2.26
## # cells per guide: median = 669 ; min = 68 ; max = 1386
## # cells per target: avg = 1390 ; min = 340 ; max = 57359
## # guides per cell: moi (avg) = 3.130242 ; min = 1 ; max = 22
## In total: 377 pos. ctrl pairs and 404000 neg. ctrl pairs.
##      pass_qc_75 pass_qc_trt pass_qc_def
## FALSE      298866      12262      92596
## TRUE       105511      392115      311781
```



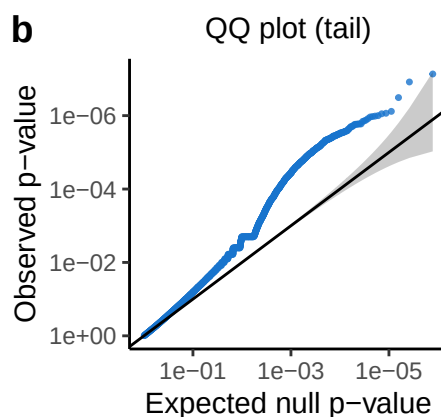
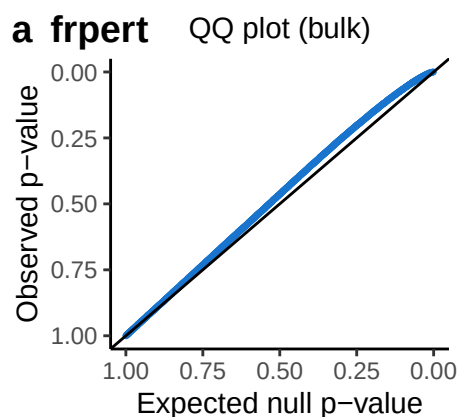
QC: pass_qc_75
Num. signif
(at alpha 0.1):
3 of 105141



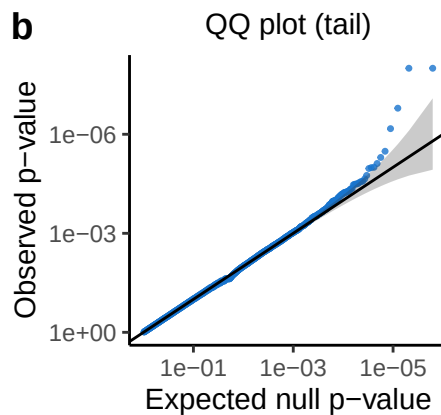
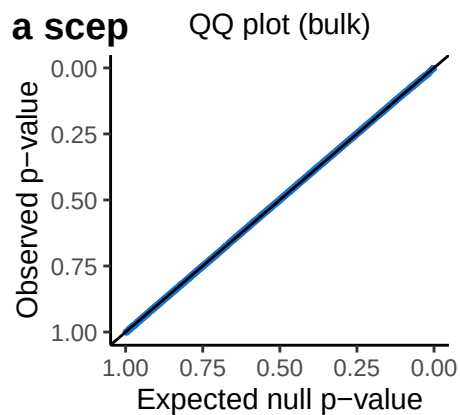
QC: pass_qc_75
Num. signif
(at alpha 0.1):
309 of 105141



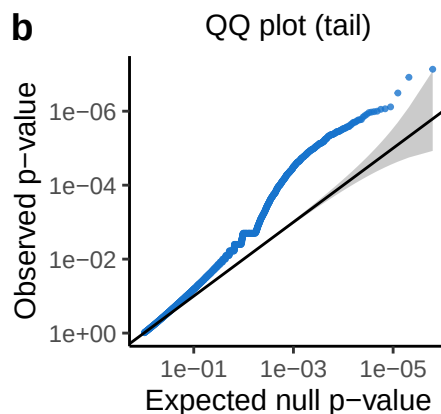
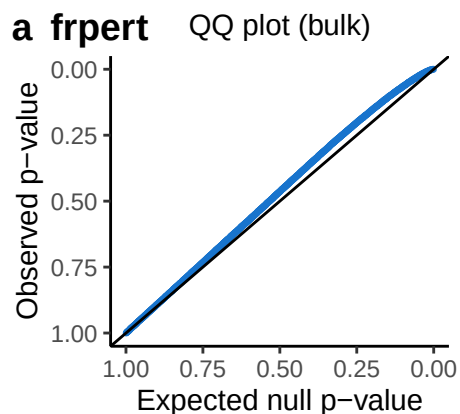
QC: pass_qc_trt
Num. signif
(at alpha 0.1):
4 of 391738



QC: pass_qc_trt
Num. signif
(at alpha 0.1):
1201 of 391738

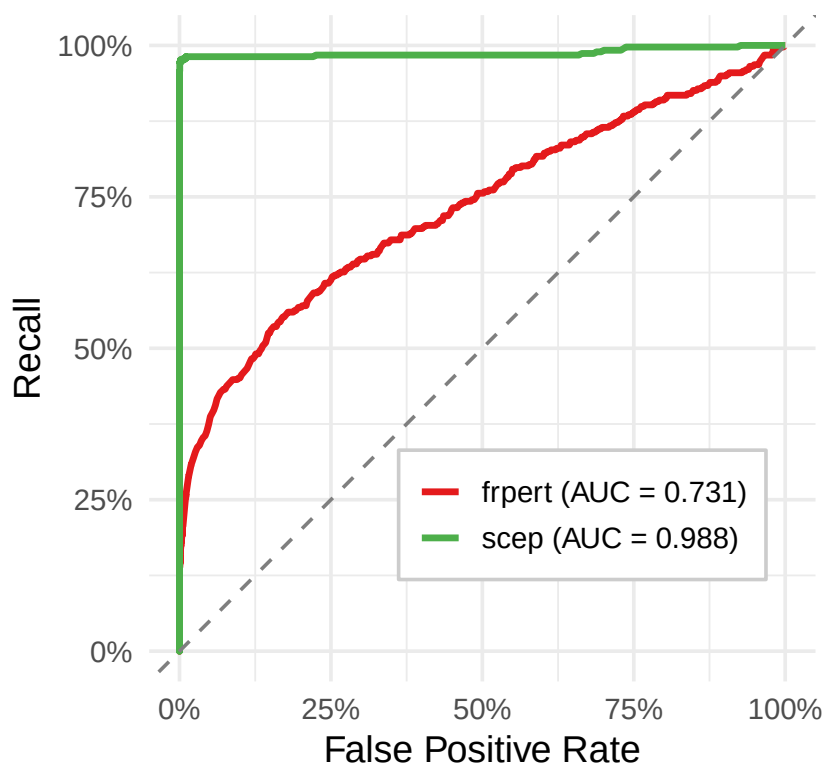


QC: pass_qc_def
 Num. signif
 (at alpha 0.1):
 4 of 311404



QC: pass_qc_def
 Num. signif
 (at alpha 0.1):
 998 of 311404

Separation of pos. and neg. (QC = pass_qc_trt)



Separation of pos. and neg. (QC = pass_qc_75)

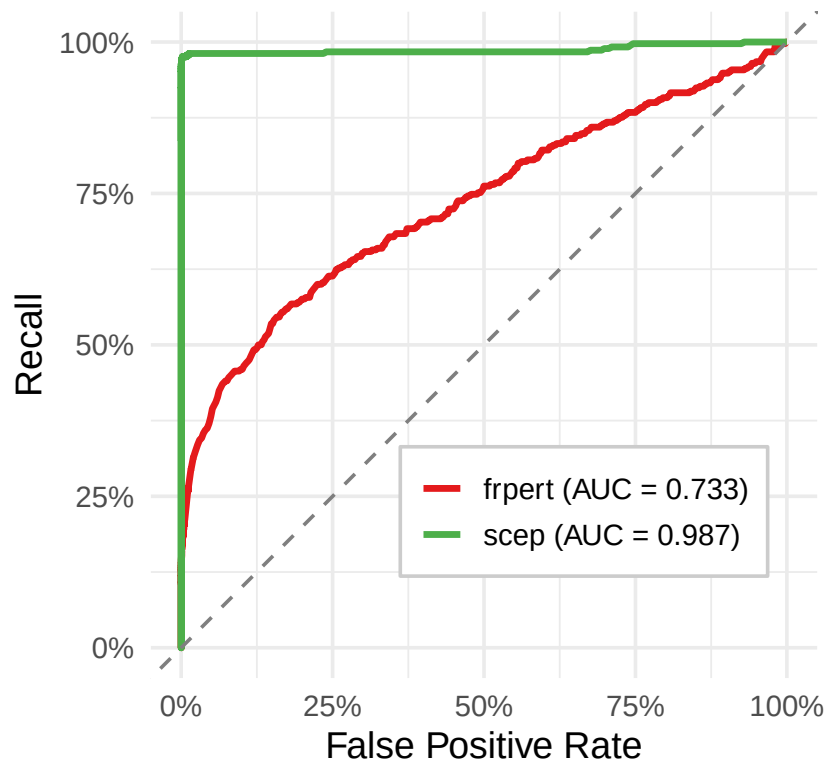
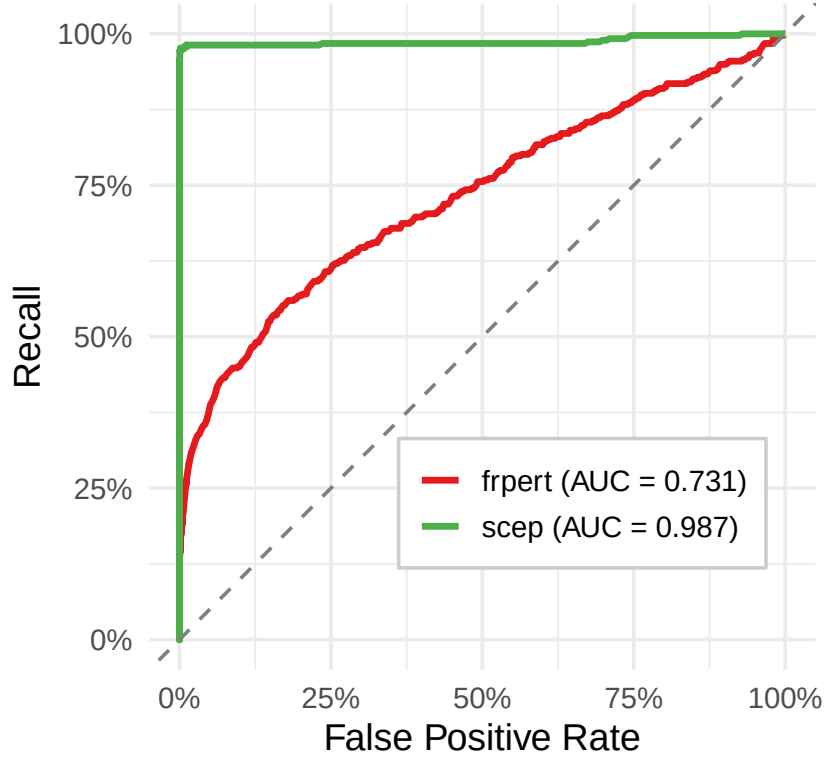
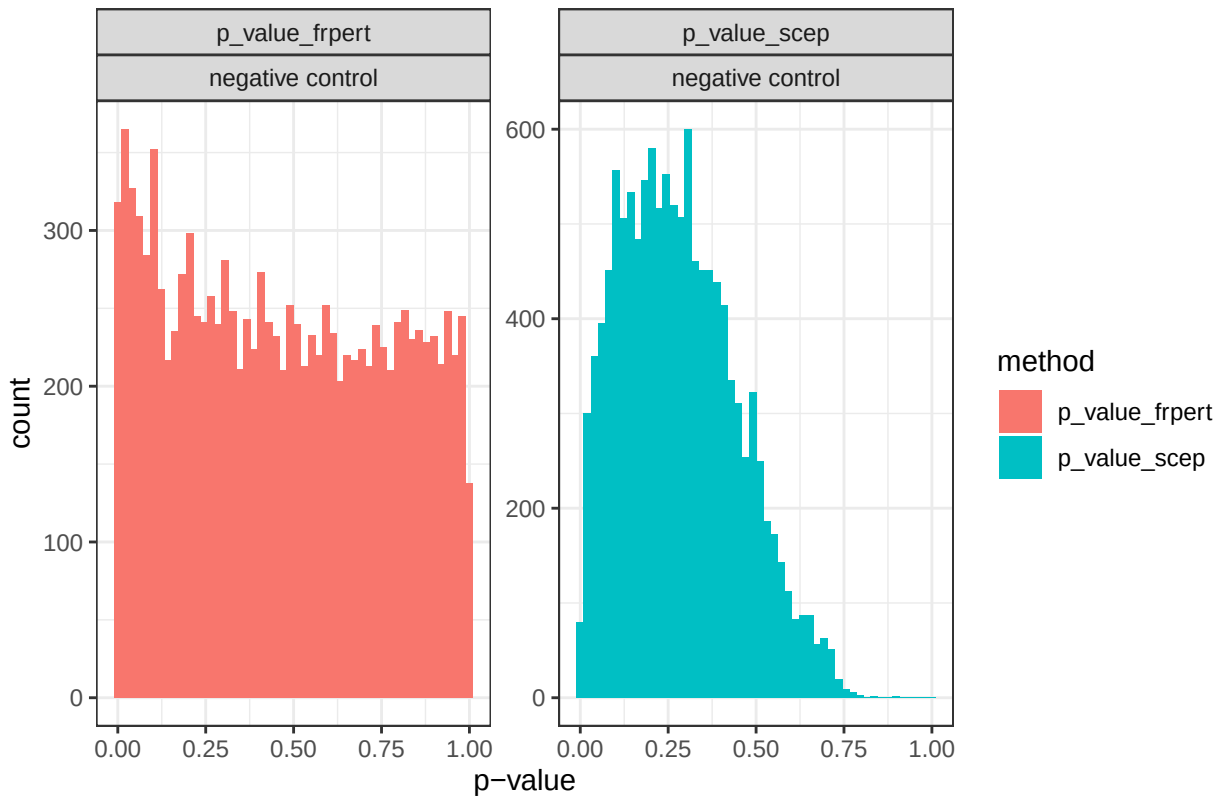


Figure 1: ROC analysis

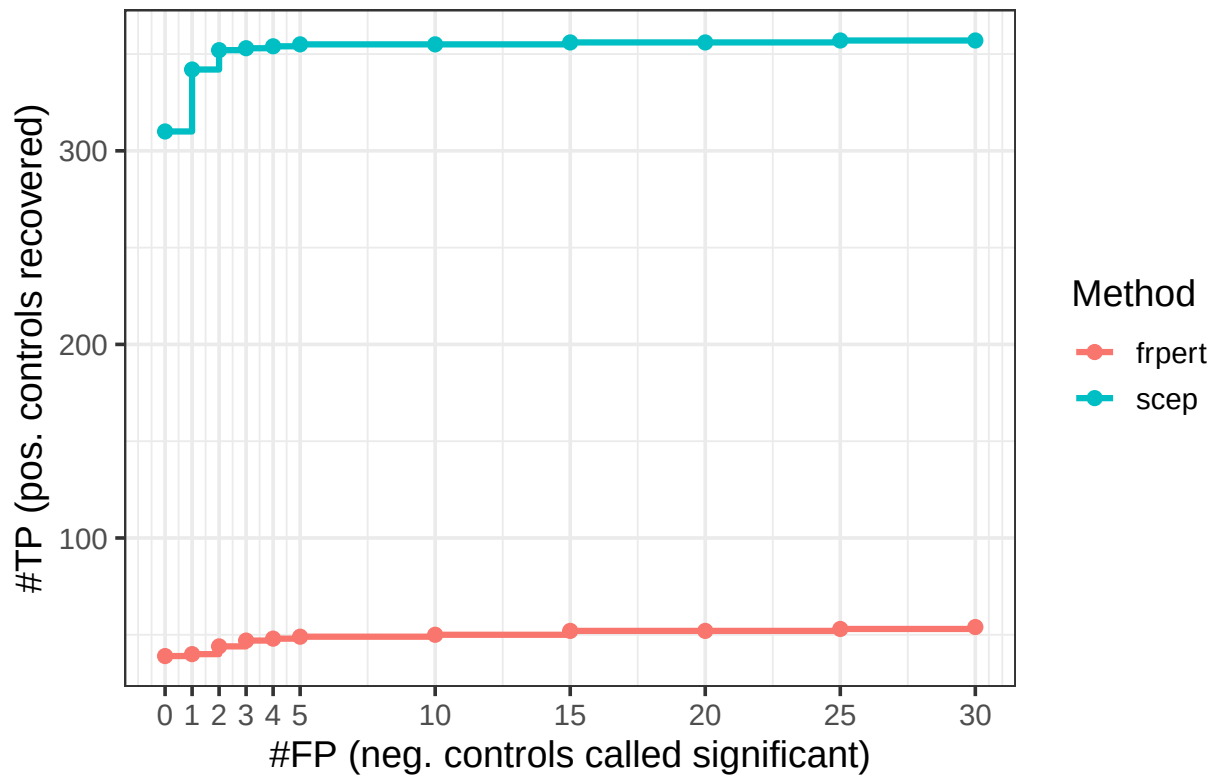
Separation of pos. and neg. (QC = pass_qc_def)



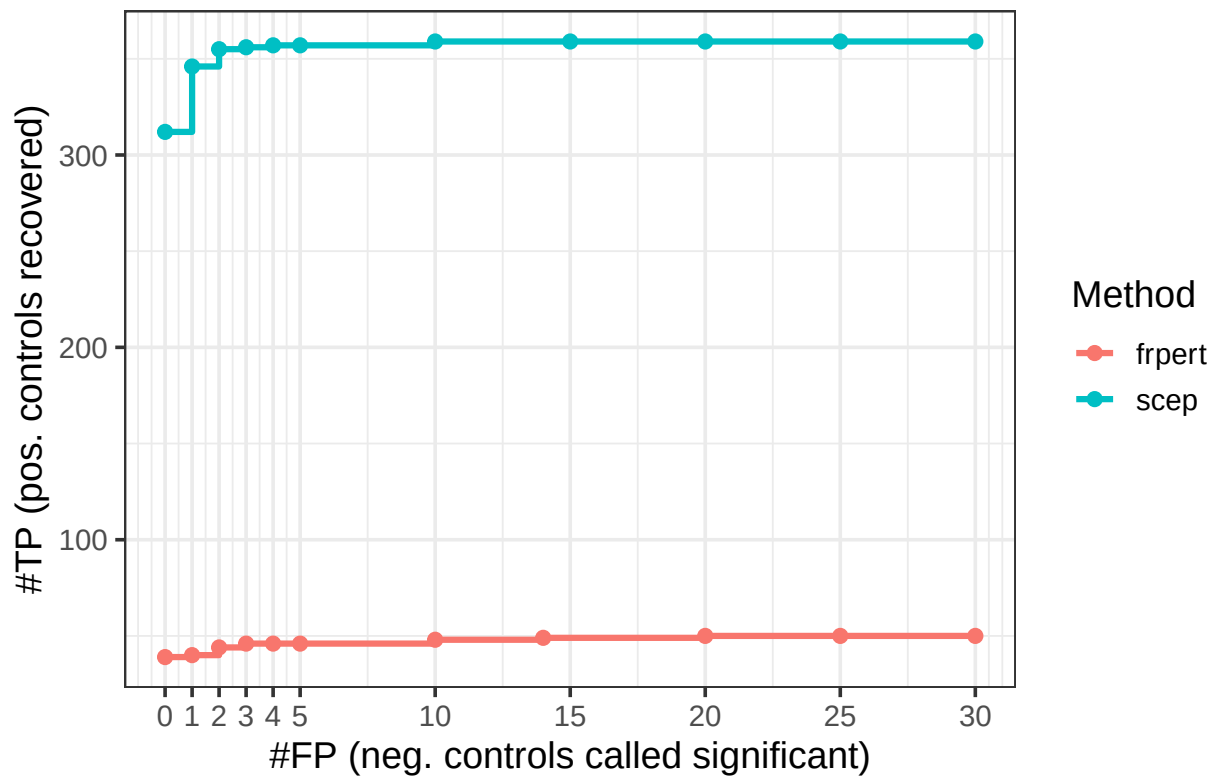
All pairs with n_nonzero_trt = 0 (no QC)



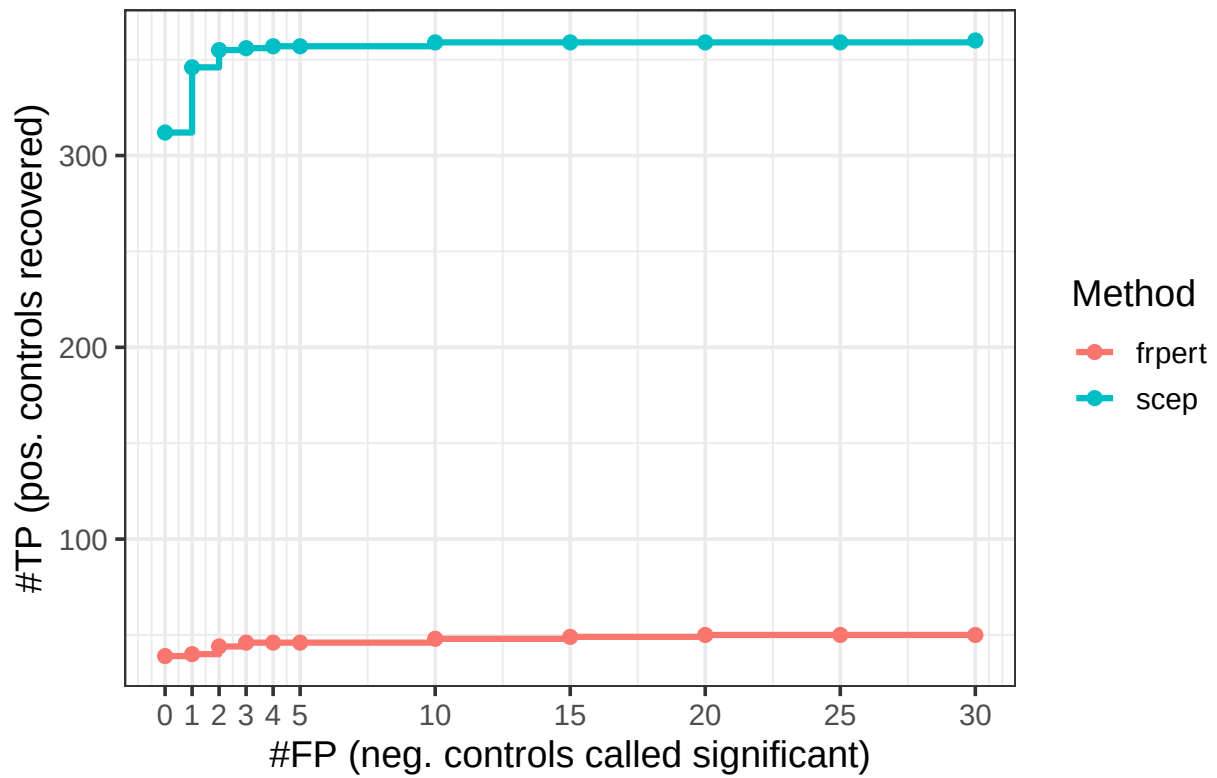
#TP recovered vs #FP allowed (QC = pass_qc_75)



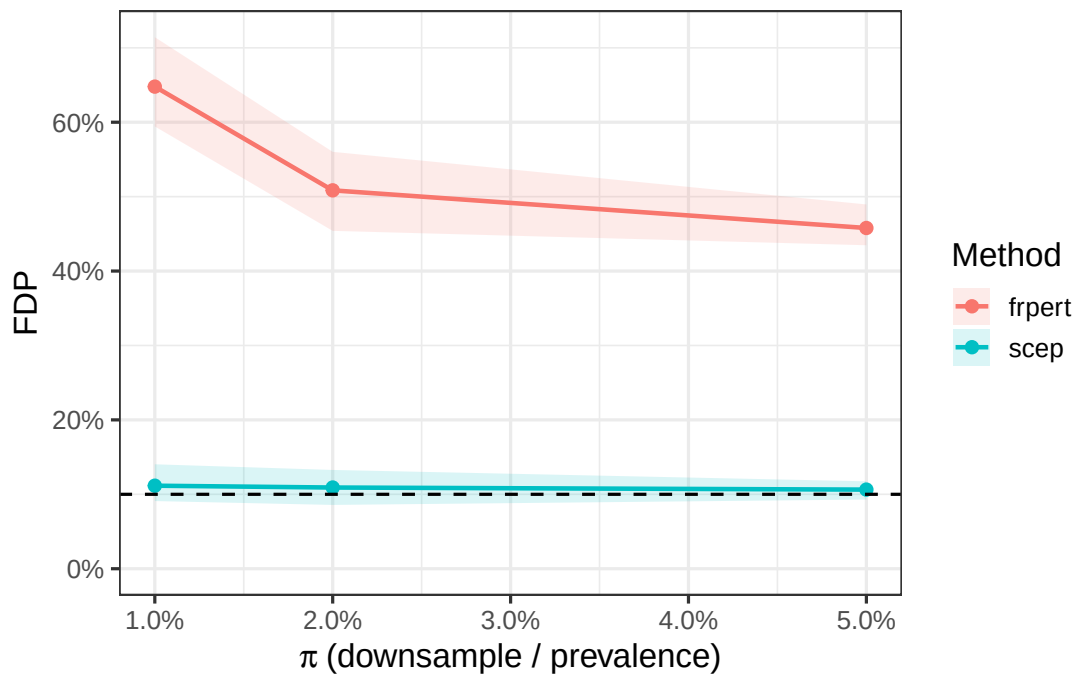
#TP recovered vs #FP allowed (QC = pass_qc_trt)



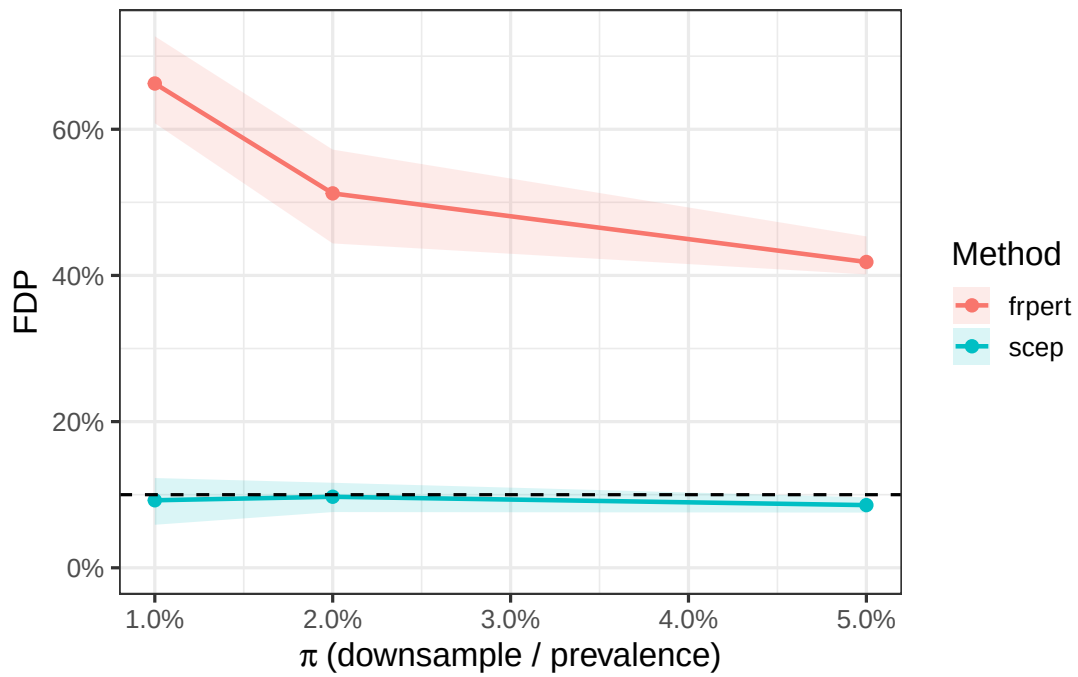
#TP recovered vs #FP allowed (QC = pass_qc_def)



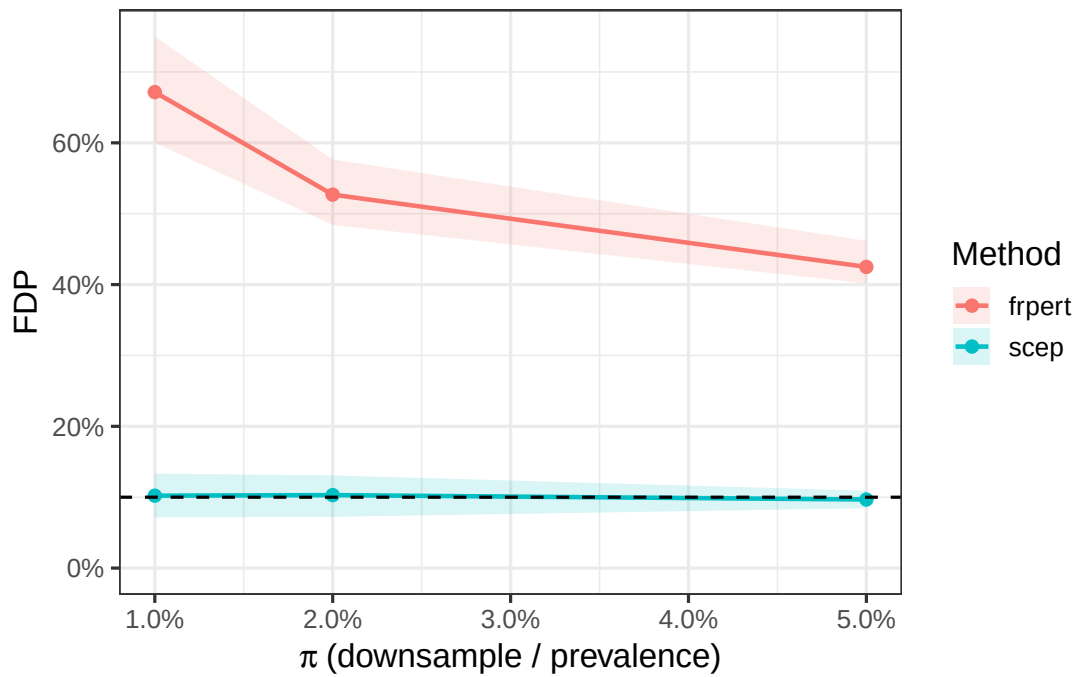
BH q = 0.1 (N=5000, B=100, QC = pass_qc_75)



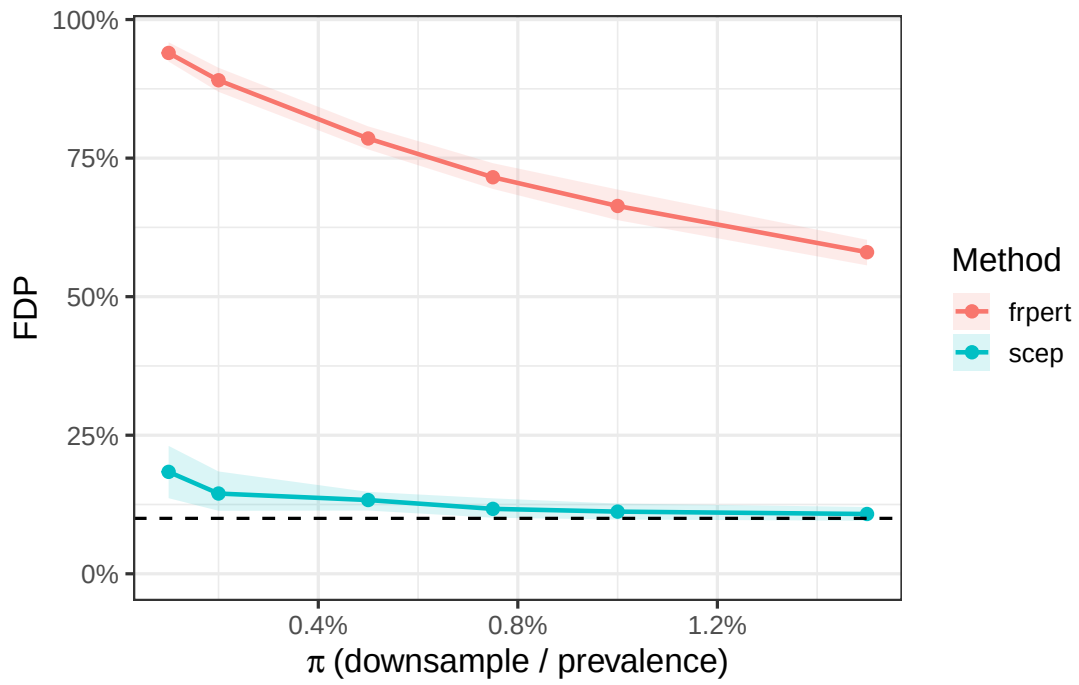
BH $q = 0.1$ (N=5000, B=100, QC = pass_qc_trt)



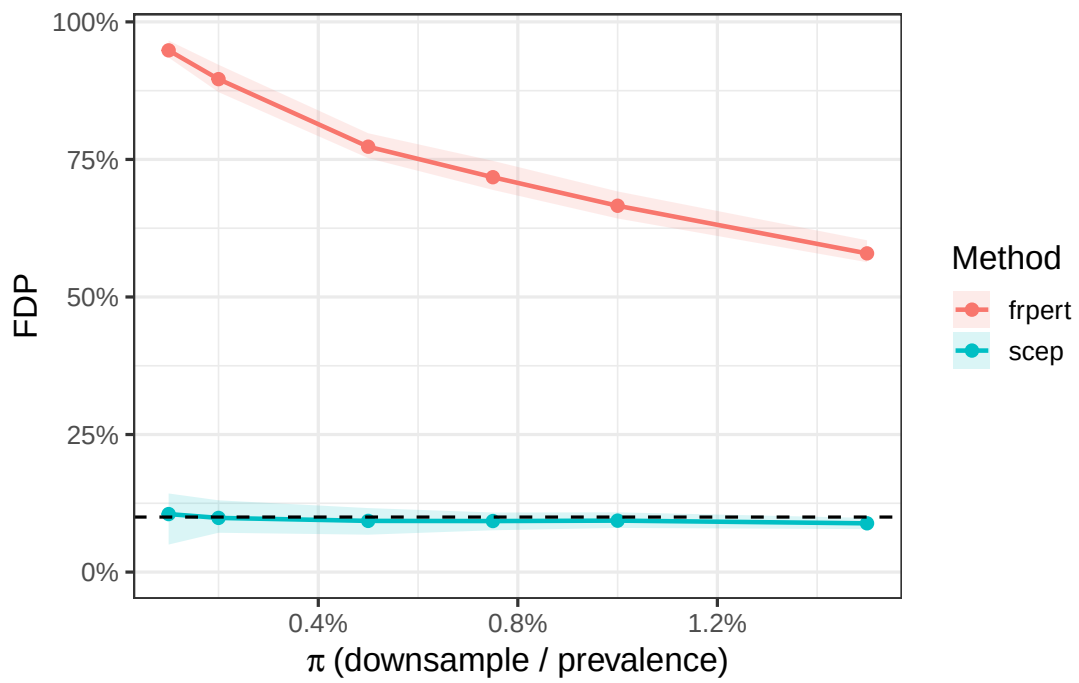
BH $q = 0.1$ (N=4000, B=100, QC = pass_qc_def)



BH $q = 0.1$ (N=20000, B=100, QC = pass_qc_75)



BH $q = 0.1$ (N=20000, B=100, QC = pass_qc_trt)



BH $q = 0.1$ (N=20000, B=100, QC = pass_qc_def)

